# Digital Medical Education Empowered by Intelligent Fabric Space

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**Abstract**

Medical education plays an important role in promoting the development of global medical science. Nevertheless, the intrinsic gap existing between institutional medical teaching and practical clinical tasks causes low education efficiency and students' weak initiative. Recent developments of sensing fabric and embedded computing, along with the advances in Artificial intelligence (AI) and digital twin technology are paving the way for the transformation of medical research towards digitization. In this work, we present an intelligent fabric space based on novel functional fabric materials and digital twin networking enabled by 5G and Internet of Things (IoT) technologies. In this space, medical students can learn knowledge with collaborative mapping of the digital and real world, cyber-physical interaction and real-time tactile feedback. And the proposed service system will evaluate and feedback students' operational behaviors to improve their experimental skills. We provide four typical applications of intelligent fabric space for medical education, including medical education training, health and behavior tracking, operation playback and reproduction, as well as medical knowledge popularization. The proposed intelligent fabric space has the potential to promote innovative technologies for training cutting-edge medical students by effective and efficient ways.